LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the core syllabus sentence (principal values), the analysis of composite inverse-trig functions — piecewise closed forms, parity/period, graph shapes, extrema on intervals — is standard olympiad and coaching-extension material. Every model here is **derivable from Part I** (branch reads + addition formulas): no new theory is assumed, only sustained bookkeeping. Nothing here "can never appear" in JEE Advanced; it is simply less typical than the Part I types.

The full closed form of f(x) = sin⁻¹(sin x) + cos⁻¹(cos x)

Both pieces are sheet-02 reads. Reduce x to $t \in [0, 2\pi)$ first (both components have period 2π, so f has period 2π). On $[0, 2\pi)$:

$$\sin^{-1}(\sin t) = t \ (0 \le t \le \tfrac{\pi}{2}), \qquad = \pi - t \ (\tfrac{\pi}{2} < t < \tfrac{3\pi}{2}),$$

$$= t - 2\pi \ (\tfrac{3\pi}{2} \le t < 2\pi)$$

$$\cos^{-1}(\cos t) = t \ (0 \le t \le \pi), \qquad = 2\pi - t \ (\pi < t < 2\pi)$$

Adding on each sub-interval of $[0, 2\pi)$ — leg by leg (the composition of two affine pieces is affine, so only endpoints need checking):

- $t \in [0, \frac{\pi}{2}]$: $t + t = 2t$ (rises $0 \to \pi$).
- $t \in [\frac{\pi}{2}, \pi]$: $(\pi - t) + t = \pi$ (the plateau).
- $t \in [\pi, \frac{3\pi}{2}]$: $(\pi - t) + (2\pi - t) = 3\pi - 2t$ (falls $\pi \to 0$).
- $t \in [\frac{3\pi}{2}, 2\pi)$: $(t - 2\pi) + (2\pi - t) = 0$ (the floor).

$$f(t) = 2t \text{ on } [0, \tfrac{\pi}{2}], \qquad f(t) = \pi \text{ on } [\tfrac{\pi}{2}, \pi]$$

$$f(t) = 3\pi - 2t \text{ on } [\pi, \tfrac{3\pi}{2}], \qquad f(t) = 0 \text{ on } [\tfrac{3\pi}{2}, 2\pi)$$

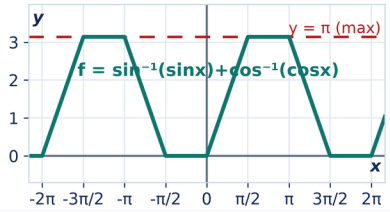
Range on $\mathbb{R}$: f takes every value in $[0, \pi]$ (max π on the plateau, min 0). **Neither odd nor even**; $f(x + 2\pi) = f(x)$.

SHORT DERIVATION

The bookkeeping lesson — always split at the branch knots

The knots of $\sin^{-1}(\sin x)$ are $x = \frac{\pi}{2} + k\pi$ (where the sawtooth turns); those of $\cos^{-1}(\cos x)$ are $x = k\pi$. A sum's piecewise form splits at the **union** of knots. On each open piece both components are linear ($\pm x + \text{const}$), so f is affine there: the whole problem reduces to assembling slopes. Slopes of $\sin^{-1}(\sin x)$ alternate $+1, -1$; of $\cos^{-1}(\cos x)$: $+1, -1$; sums have slopes in $\{-2, 0, 2\}$ — the flat middle leg above is the $+1 + (-1)$ collision.

f(x): plateau π on [π/2, π], zero on [3π/2, 2π), period 2π — range [0, π]



Sibling composites worth owning

- $g(x) = \sin^{-1}(\sin x)$ *alone: odd, 2π-periodic, range $[-\pi/2, \pi/2]$.*
- $h(x) = \cos^{-1}(\cos x)$*: even, 2π-periodic, range $[0, \pi]$ — note $h(x) = |x|$ on $[-\pi, \pi]$, the "triangle wave".*
- $p(x) = \tan^{-1}(\tan x)$*: odd, π-periodic (undefined at odd multiples of $\pi/2$), range $(-\pi/2, \pi/2)$ — the sawtooth of sheet 02.*
- $q(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x)$*: collapses to $\frac{\pi}{2}$* ***everywhere*** *(complementary pair with the same argument $\sin x$ — a one-liner that looks like work).*
- $r(x) = \tan^{-1}(\sin x)$*: odd, 2π-periodic, range $[-\frac{\pi}{4}, \frac{\pi}{4}]$ (bound $\sin x \in [-1, 1]$ through the increasing $\tan^{-1}$). Extrema-type questions ("range of $\tan^{-1}(\sin x) + \tan^{-1}(\cos x)$") are solved by monotone-plus-derivative-free bounding: $\sin x + \cos x$-style collapse.*

STANDARD APPLICATION

Extrema method for
$u(x) = \tan^{-1}(\sin x) + \tan^{-1}(\cos x)$
Add via the table:
$\tan^{-1} u + \tan^{-1} v = \tan^{-1} \frac{u+v}{1-uv}$ *when* $uv < 1$ *— here* $uv = \sin x \cos x \le \frac{1}{2} < 1$ ✓ *always. So*

$$u(x) = \tan^{-1}\left(\frac{\sin x + \cos x}{1 - \sin x \cos x}\right)$$

a monotone transform of $s = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$ *with* $\sin x \cos x = \frac{s^2-1}{2}$*. Write* $u = \tan^{-1} \frac{s}{1 - \frac{s^2-1}{2}} = \tan^{-1} \frac{2s}{3-s^2}$*. On* $s \in [-\sqrt{2}, \sqrt{2}]$ *the fraction* $\frac{2s}{3-s^2}$ *is odd-increasing in s (denominator $3 - s^2 \in [1, 3]$ stays positive), so max at* $s = \sqrt{2}$*:* $\frac{2\sqrt{2}}{1} = 2\sqrt{2}$*, i.e.* $\max u = \tan^{-1}(2\sqrt{2})$ *and* $\min u = -\tan^{-1}(2\sqrt{2})$*; range* $[-\tan^{-1} 2\sqrt{2}, \tan^{-1} 2\sqrt{2}]$*. Every step used only Part I: the addition formula's condition, the range of $\sin x + \cos x$, monotonicity of $\tan^{-1}$.*

COMMON TRAP

Traps specific to piecewise work

- *Splitting at the wrong knots (only one component's) creates false affine pieces with wrong slopes.*
- *Period: $\sin^{-1}(\sin x) + \cos^{-1}(\cos x)$ has period 2π; $\tan^{-1}(\tan x)$ has π — sums take the LCM (2π here).*
- *Endpoints of branch windows ($x = \frac{\pi}{2}$ etc.) are continuous for these composites but not differentiable — extrema questions may land on a knot; evaluate both one-sided values (they agree here).*
- *In "$q(x)$ collapses to $\pi/2$" the argument of both inverses must be identical; $\sin^{-1}(\sin x) + \cos^{-1}(\cos x) \neq \frac{\pi}{2}$ (that is f above!).*

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO standard)

Telescoping sums of $\tan^{-1}$-terms are a staple of IOQM/RMO arithmetic. The engine is Part I's difference formula; the skill is recognising which $\frac{a-b}{1+ab}$ is hiding inside the term. Sums are treated exactly (finite forms and the convergent infinite limits); no calculus beyond "$\tan^{-1} n \to \frac{\pi}{2}$ as $n \to \infty$" is used (that limit is the graph of $\tan^{-1}$ from Part I).

The one identity every such sum hides

KEY RESULT

Difference identity (the telescoping workhorse)

For $ab > -1$:

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1}\left(\frac{a-b}{1+ab}\right)$$

So any term $\tan^{-1}\frac{c}{d}$ is telescoping material the moment $c = a-b$ and $d = 1+ab$ for an integer-valued sequence a_k, b_k (usually $b_k = a_{k+1}$ or $a_k = k + \text{const}$). With $a = k+1$, $b = k$: $\frac{a-b}{1+ab} = \frac{1}{1+k+k^2}$; with $a = k+1$, $b = k-1$: $\frac{2}{k^2}$. Those two shapes cover most published problems.

SHORT DERIVATION

Series 1 — $\sum_{k=1}^{n} \tan^{-1}\frac{1}{1+k+k^2} = \tan^{-1}(n+1) - \frac{\pi}{4}$

Each term:
$\frac{1}{1+k(k+1)} = \frac{(k+1)-k}{1+k(k+1)} = \tan^{-1}(k+1) - \tan^{-1}$
(valid: $k(k+1) > -1$ ✓). The sum telescopes:

$$\sum_{k=1}^{n} [\tan^{-1}(k+1) - \tan^{-1} k]$$

$$\tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1}(n+1) - \tfrac{\pi}{4}.$$

Checks: $n = 1$: $LHS = \tan^{-1}\frac{1}{3} \approx 0.3217$;
$RHS = \tan^{-1} 2 - \frac{\pi}{4} \approx 1.1071 - 0.7854 = 0.3217$ ✓. As $n \to \infty$: sum $\to \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

SHORT DERIVATION

Series 2 — $\sum_{k=1}^{n} \tan^{-1}\frac{2}{k^2} = \tan^{-1}(n+1) + \tan^{-1} n - \frac{\pi}{4}$

Here $\frac{2}{k^2} = \frac{(k+1)-(k-1)}{1+(k+1)(k-1)}$, so
$\tan^{-1}\frac{2}{k^2} = \tan^{-1}(k+1) - \tan^{-1}(k-1)$
(valid since $(k+1)(k-1) = k^2 - 1 > -1$).
This is a **two-step** telescope (stride 2):

$$\sum_{k=1}^{n} [\tan^{-1}(k+1) - \tan^{-1}(k-1)]$$

$$[\tan^{-1}(n+1) + \tan^{-1} n] - [\tan^{-1} 0 + \tan^{-1} 1]$$

$$= \tan^{-1}(n+1) + \tan^{-1} n - \tfrac{\pi}{4}.$$

Checks: $n = 1$: $LHS = \tan^{-1} 2 \approx 1.1071$;
$RHS = \tan^{-1} 2 + \tan^{-1} 1 - \frac{\pi}{4} = 1.1071$ ✓.
Infinite limit: $\frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}$ — the classic "$\sum_{k=1}^{\infty} \tan^{-1}\frac{2}{k^2} = \frac{3\pi}{4}$".

Worked variants — the same engine, different residues

WORKED EXAMPLE

Variant A — $\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$ (the denominator is the tell)

Spot $a = k^2+k+1$, $b = k^2-k+1$: then $a-b = 2k$ ✓ and, from $(k^2+1)^2 - k^2 = k^4+k^2+1$,

$$1+ab = 1+(k^2+k+1)(k^2-k+1)$$

$$= k^4+k^2+2$$

exactly the given denominator. Each term is $\tan^{-1}(k^2+k+1) - \tan^{-1}(k^2-k+1)$, and

$k^2-k+1 = (k-1)^2+(k-1)+1 = u_{k-1}$ with $u_k = k^2+k+1$ — the sum telescopes along u_k:

$$\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$$

$$= \tan^{-1}(n^2+n+1) - \tan^{-1}1$$

$$= \tan^{-1}(n^2+n+1) - \frac{\pi}{4}.$$

Numeric verification (also in `math_checks.py`): $n=1$: $\tan^{-1}\frac{2}{4} = 0.46365 = \tan^{-1}3 - \frac{\pi}{4}$ ✓. $n=2$: $\tan^{-1}0.5 + \tan^{-1}\frac{2}{11} = 0.64350 = \tan^{-1}7 -$ ✓.

STANDARD APPLICATION

Strategy card for arctan series

1. Force the term into the shape $\tan^{-1}\frac{a-b}{1+ab}$ by **matching the denominator to** $1+ab$ with small integer a_k, b_k.
2. Ensure $a_k b_k > -1$ on the range (automatic for positive sequences).
3. Telescoping: write out the first two and last two terms before cancelling; stride-2 series lose half their terms to the boundary.
4. Infinite sums: take $n \to \infty$ in the closed form, using $\tan^{-1} n \to \frac{\pi}{2}$; never sum "infinitely many small arctans" by estimation.
5. Verify $n=1$ and $n=2$ numerically before trusting the closed form.

COMMON TRAP

Traps — series

- Stride-2 telescopes drop the parity of surviving boundary terms; write n even and $n=5$ by hand and compare.
- $\tan^{-1}\frac{2}{k^2}$ at $k=1$: $\tan^{-1}2$; the split $\tan^{-1}2 - \tan^{-1}0$ is legal ($0\cdot 2 = 0 > -1$ ✓) — but at $ab=-1$ the difference formula is void: e.g. $a=1, b=-1$ gives $\tan^{-1}1 - \tan^{-1}(-1) = \frac{\pi}{2}$ while $\frac{a-b}{1+ab} = \frac{2}{0}$ explodes — the boundary case is real, not a formality.
- Sign of the residue: $\tan^{-1}\frac{1}{1+k+k^2}$ telescopes as $\tan^{-1}(k+1) - \tan^{-1}k$ (positive), not $k-(k+1)$.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO + coaching classics)

These are "prove/evaluate with structure" problems: symmetric conditions on x, y, z forcing angle sums, nested-inverse expressions, and range problems where the branch constraints are the hidden system of inequalities. All are derivable from Part I; the added value is problem-recognition. No claim is made that such items "can never appear" in JEE Advanced — they are simply the harder end of the same toolkit.

The classic: x + y + z = xyz forces A + B + C = π

KEY RESULT

Theorem (with its hypothesis)

If $x, y, z > 0$ and $x + y + z = xyz$, then

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi.$$

More symmetrically: with $A = \tan^{-1} x$ etc. all in $(0, \frac{\pi}{2})$, the condition $x + y + z = xyz$ is exactly $\tan(A + B + C) = 0$ — and the range bookkeeping ($A + B + C \in (0, \frac{3\pi}{2})$) pins $A + B + C = \pi$ (it cannot be 0: each angle is positive).

SHORT DERIVATION

Proof — expand $\tan(A + B + C)$

With $A = \tan^{-1} x, B = \tan^{-1} y, C = \tan^{-1} z$:

$$\tan(A + B + C) = \frac{x + y + z - xyz}{1 - (xy + yz + zx)}.$$

Hypothesis: the numerator vanishes. So either the denominator is nonzero and $\tan(A + B + C) = 0$, giving $A + B + C = k\pi$; since $A + B + C \in (0, \frac{3\pi}{2})$ the only candidate is π. Or the denominator vanishes too ($xy + yz + zx = 1$): then $A + B + C \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ — one checks directly that $x, y, z > 0$ with $x + y + z = xyz$ forces $xy + yz + zx > 1$, so this degenerate case cannot occur (e.g. by AM-GM: $xyz = x + y + z \geq 3\sqrt[3]{xyz}$ gives $xyz \geq 3\sqrt{3}$, while $xy + yz + zx \geq 3(xyz)^{2/3} \geq 9 > 1$). ∎

Sanity: $x = y = z = \sqrt{3}$: sum $= 3\sqrt{3}$, product $= 3\sqrt{3}$ ✓, and $3\tan^{-1}\sqrt{3} = 3 \cdot \frac{\pi}{3} = \pi$ ✓.

WORKED EXAMPLE

Companion identity — $xy + yz + zx = 1$ forces the sum to $\frac{\pi}{2}$

If $x, y > 0$ and $xy < 1$ then $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$; with $x, y, z > 0$ and $xy + yz + zx = 1$, pairwise: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ and $1 - xy = z(x + y)$, so $\frac{x+y}{1-xy} = \frac{1}{z}$ and $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{1}{z} = \frac{\pi}{2} - \tan^{-1} z$ (complementary pair!). Hence

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$$

whenever $x, y, z > 0$, $xy + yz + zx = 1$.

(Condition $xy < 1$ is automatic: $xy < xy + yz + zx = 1$ ✓.) Example: $x = y = \frac{1}{2}$, then $z = \frac{1 - 1/4}{1} = \frac{3}{4}$... check: $xy + yz + zx = \frac{1}{4} + \frac{3}{8} + \frac{3}{8} = 1$ ✓ and $2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{3}{4} \approx 0.9273 + 0.6435 = 1.5$ ✓.

Nested and mixed constraint problems

Nested and mixed constraint problems

STANDARD APPLICATION

Pattern 1 — nested arctans: $\tan^{-1}(\frac{1}{2})+\tan^{-1}(\frac{1}{5})+\tan^{-1}(\frac{1}{8})$-style chains

Add two at a time with the $xy<1$ face (small positive arguments keep you in the plain case): $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}=\tan^{-1}\frac{7}{9}$ (since $\frac{1/2+1/5}{1-1/10}=\frac{7/10}{9/10}$); then $+\tan^{-1}\frac{1}{8}$: $\tan^{-1}\frac{7}{9}+\tan^{-1}\frac{1}{8}=\tan^{-1}\frac{65/72}{65/72}=\tan^{-1}1=\frac{\pi}{4}$. So the chain sums to $\frac{\pi}{4}$ exactly. The discipline: after each addition, re-check the xy condition with the *new* combined argument.

STANDARD APPLICATION

Pattern 2 — the branch constraints as a system: solve $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$ given $x+y=1$

Take the sine of both sides: $x\sqrt{1-y^2}+y\sqrt{1-x^2}=1$. With $y=1-x$ this becomes $x\sqrt{2x-x^2}+(1-x)\sqrt{2x-x^2}=\sqrt{2x-x^2}=$, so $2x-x^2=1$, $(x-1)^2=0$, $x=1, y=0$. Check: $\sin^{-1}1+\sin^{-1}0=\frac{\pi}{2}$ ✓. Alternatively read the equation itself: $\sin^{-1}y=\frac{\pi}{2}-\sin^{-1}x=\cos^{-1}x=\sin^{-1}\sqrt{1-$ needs $y=\sqrt{1-x^2}$ (branch: $y\geq 0$) — together with $x+y=1$: $x+\sqrt{1-x^2}=1$ gives $x\in\{0,1\}$, and both $(1,0)$ and $(0,1)$ work by symmetry. **Two solutions** — a squaring route may lose one; the branch route sees both.

STANDARD APPLICATION

Pattern 3 — bounding an arctan sum without computing it

For positive a_i, $\tan^{-1}$ is increasing and concave on $\mathbb{R}^+$, but the exam-legitimate bound is cruder: $\tan^{-1}x<\frac{\pi}{2}$ and $\tan^{-1}x\leq x$ for $x\geq 0$ (from the graph: tangent line at 0 is $y=x$, curve below it). So e.g. $\sum_{k=1}^{100}\tan^{-1}\frac{1}{k^2}<\sum_{k=1}^{100}\frac{1}{k^2}<\frac{\pi^2}{6}<2$, and since also each term >0 the sum is trapped in $(0,2)$; sharper telescoping (oly-02) gives its exact value when the shape allows. Bounds + exact-when-possible is the olympiad habit.

COMMON TRAP

Traps — structure problems

- $x+y+z=xyz\Rightarrow$ sum $=\pi$ **requires positivity** (x,y,z of mixed signs can give 0 or $-\pi$; e.g. $x=y=z=0$ satisfies the equation with sum 0).
- $xy+yz+zx=1$ identity uses the complementary pair $\tan^{-1}\frac{1}{z}=\frac{\pi}{2}-\tan^{-1}z$ — true because $\frac{1}{z}>0$; for $z<0$ it fails (both sides off by π-jumps).
- In constraint systems, the range of each inverse is a real inequality: $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$ already forces $x,y\geq 0$ (LHS $\geq -\pi$, but for the sum to reach $\frac{\pi}{2}$ with both terms $\leq\frac{\pi}{2}$ and $\geq -\frac{\pi}{2}$ you need… precisely: $x=\sin A, y=\sin B$ with $A+B=\frac{\pi}{2}$, $A,B\in[-\frac{\pi}{2},\frac{\pi}{2}]$ forces $A,B\in[0,\frac{\pi}{2}]$, so $x,y\geq 0$ and $y=\cos A=\sqrt{1-x^2}$: the solution set is exactly the quarter circle $x^2+y^2=1$, $x,y\geq 0$ — the constraint $x+y=1$ then intersects it in $\{(1,0),(0,1)\}$ as computed).
- Two-step telescopes and nested arctans: re-check the addition-formula condition after **every** combination; a chain is only as safe as its last link.